

Customer Behavior & Subscription Propensity

Predictive analytics turns behavior into a conversion pipeline

01

3,900

Customer shopping records analyzed

02

Subscription likelihood

Classification models score individual subscription likelihood

03

Prioritized growth targets

Output goes beyond descriptive reporting

The project connects four layers from raw data to action



High-intent non-subscribers are a small, valuable audience



Test cohort: 780 customers

45

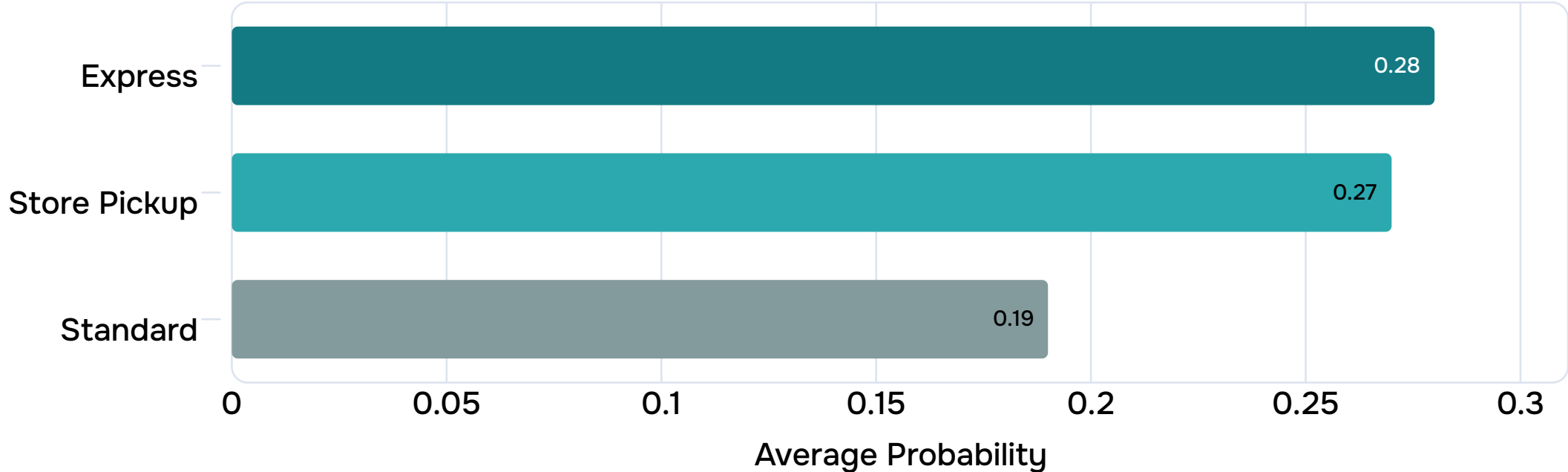
High-propensity users

5.77% of the test cohort

Focus outreach on precision, not broad discounting

Shipping choice is a visible signal of subscription intent

Shipping Type



Average subscription probability, approximately $\cdot \text{Express Shipping} / \text{Store Pickup} / \text{Standard Shipping}$

Use fulfillment behavior to sharpen campaign prioritization

High-propensity customers already behave like loyal buyers

~28.38

Average previous purchases among high-propensity
customers

Relationship depth and repeat value

Frequent purchasing signals
relationship depth and repeat value

An activation opportunity

The conversion gap is an activation
opportunity, not an awareness
problem

Model performance supports operational use

Random Forest and Logistic Regression

80/20

Stratified training / test split

780

Records in the test sample

0.90–0.91

Approximate ROC-AUC

Feature importance and model coefficients expose behavioral drivers

SQL makes the model usable by growth teams

01

Filter

Non-subscribers in the High propensity tier

02

Sort

Subscription probability creates a prioritized outreach list

03

Aggregate

Average probability by shipping type guides targeting strategy

Recommended activation playbook

- 01 Target high-propensity non-subscribers first
- 02 Use purchase frequency to prioritize loyal customers
- 03 Tailor messaging around convenience and fulfillment preferences
- 04 Measure conversion lift by propensity tier and shipping behavior

From dashboard insight to measurable growth

Use propensity tiers as a repeatable operating rhythm:

Identify

Prioritize

Activate

Learn